

SOFT SKILLS FOR AI ENGINEER

CAPSTONE IDEA GENERATION: PITCHING, DOCUMENTING, AND DEFENDING IDEA

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Project Title: AI-Based Water Distribution Demand Prediction System

TASK 1: CAPSTONE IDEA BRAINSTORM

Idea 1: AI-Based Water Distribution Demand Prediction System

Problem

Urban water suppliers may experience periods when some areas require more water than others. During high-demand periods, poor demand estimation can contribute to water shortages, uneven distribution, inefficient use of available water, and difficulty planning water supply.

Water demand can change according to the day of the week, season, weather conditions, population, number of water connections, holidays, and previous demand patterns. Without reliable demand predictions, water distribution planning may depend heavily on historical experience and manual estimates.

Who has the problem

Water utilities, water distribution planners, and urban communities that depend on reliable water supply.

How they deal with it today

Water distribution decisions can rely on historical water demand records, fixed schedules, manual observations, previous experience, and estimates of expected demand.

Why AI/ML helps

A rule-based system could use fixed thresholds or historical averages, but water demand can change according to several interacting factors. Machine learning can identify patterns in historical water demand data and predict future demand for selected urban zones.

These predictions can help water utilities understand which areas are likely to require more or less water and support better distribution planning.

Possible data

The system could use historical water demand records, dates, and urban zones, number of water connections, population estimates, rainfall, temperature, previous supply levels, and information about holidays or special events.

Potential data sources could include public water datasets, weather datasets, population data, or anonymized records from a water utility. The main uncertainty is whether sufficiently detailed and reliable Rwanda-specific water demand data will be available.

Idea 2: AI-Based Electricity Consumption Forecasting and Energy Planning System

Problem

Electricity consumption changes throughout the day and can also vary according to the season, weather, holidays, and consumer behavior. Electricity providers may face difficulties planning energy supply when future consumption is not accurately estimated.

Who has the problem

Electricity providers, energy planners, businesses, and households.

How they deal with it today

Energy planning may rely on historical electricity consumption records, fixed schedules, previous experience, and manual estimates.

Why AI/ML helps

Machine learning can analyze historical electricity consumption and identify patterns related to time, weather, season, and previous usage. The system can forecast future electricity consumption and help energy planners prepare for periods of high or low demand.

Possible data

Possible data could include historical electricity consumption, date and time, temperature, rainfall, season, holidays, and customer or location information.

Data could come from public energy datasets or anonymized electricity provider records.

Idea 3: AI-Based Waste Collection Demand Prediction and Route Planning System

Problem

Waste collection services may collect waste from some areas more frequently than necessary while other areas may experience overflowing collection points. Poor planning can lead to inefficient routes, increased fuel consumption, and delays in waste collection.

Who has the problem

Waste management companies, municipal authorities, waste collection planners, and urban communities.

How they deal with it today

Waste collection can rely on fixed schedules, previous experience, manual observations, and predetermined collection routes.

Why AI/ML helps

Waste generation can vary according to location, population, day of the week, season, holidays, and previous collection patterns. Machine learning can predict

waste collection demand for different areas and help planners determine which areas require more frequent collection.

Possible data

Possible data could include historical waste quantities, collection dates, locations, population estimates, collection frequency, day of the week, holidays, and weather information. Data could come from municipal waste records, waste management companies, or suitable public datasets.

Selected Idea

I selected AI-Based Water Distribution Demand Prediction System because it addresses a practical infrastructure problem and uses AI to support better water supply planning.

The system will predict how much water different urban zones are likely to require during a future period. These predictions can help water utilities identify areas with higher expected demand and make better distribution decisions.

Compared with the electricity project, the water project focuses on a critical resource and addresses the specific challenge of predicting demand across different urban areas. The waste collection project is also useful, but it would require reliable records of waste generation and collection activities for different areas.

The main tradeoff is data availability. Detailed water demand data for different urban zones may be difficult to obtain. I am accepting this challenge because the project addresses an important real-world resource management problem and

provides an opportunity to demonstrate how AI can support better water distribution planning.

TASK 2: PLAIN-LANGUAGE PITCH

Urban water suppliers need to make sure that enough water is available for different areas, but water demand changes over time. Some areas may require more water on certain days or during particular seasons, while other areas may require less.

My project proposes an AI-Based Water Distribution Demand Prediction System.

The system will use historical water demand data together with information such as weather, rainfall, season, day of the week, population, and number of water connections. AI will analyze these patterns and predict how much water different urban zones are likely to require.

For example, if the system predicts that Zone A will require 18,500 m³ of water tomorrow and Zone B will require 12,300 m³, water planners can use this information to understand the expected demand in each area and plan water distribution accordingly.

The system will therefore help water utilities make distribution decisions based on predicted demand instead of relying only on manual estimates or previous experience.

The main goal is to improve water demand prediction, support better water distribution, reduce the risk of shortages, and make better use of available water resources.

TASK 3: PEER FEEDBACK EXCHANGE

3.1 Feedback I Give

Genuinely compelling

I found the idea interesting because it uses AI to solve a practical problem related to water supply. I especially liked that the system predicts water demand for different urban areas and can help planners make better distribution decisions.

Risk or gap

One important risk is the availability of reliable data. The system needs historical water demand data from different areas, and additional information such as rainfall, weather, population, and water connections may not always be available.

Specific question

What would you do if you cannot find enough reliable water demand data for several urban zones? Would you reduce the number of zones or use another available dataset?

3.2 Feedback I Receive

Feedback received

My peer was concerned about whether enough reliable water demand data would be available to train and test the AI model. They also suggested that using too many urban zones could make the project difficult to complete within the available time.

What I agree with and will change

I agree that data availability is an important issue for this project. I will reduce the initial scope to a small number of selected urban zones and use only variables that can realistically be obtained.

I will also check the available datasets early in the project so that I can determine whether they are sufficient for training and evaluating the model.

What I disagree with and why

I do not think the project should be limited to predicting demand for only one area. Predicting demand across selected urban zones gives the project more practical value because it allows planners to compare expected demand between areas.

However, I will keep the number of zones limited so that the project remains realistic and achievable within the available time.

Note: The feedback in this section should be adjusted to match the actual comments received from your classmate.

TASK 4: PROJECT PROPOSAL DOCUMENTATION

1. Problem Statement and Target User

Urban water demand varies across different areas and time periods. Water suppliers may face difficulty predicting how much water each area will require because demand can change according to historical demand, weather, rainfall, season, population, number of water connections, and other factors.

Inaccurate demand estimation can contribute to inefficient water distribution and may increase the risk of shortages in areas with high demand

The target users are water utility planners, distribution managers, and authorized water supply personnel responsible for water distribution planning.

The initial system will focus on selected urban zones rather than attempting to predict water demand for an entire national distribution network.

2. Proposed Approach

The proposed system will use machine learning to predict future water distribution demand for selected urban zones.

First, the system will collect historical water demand information for the selected zones.

Relevant variables may include:

- * Historical water demand
- * Date
- * Day of the week
- * Month or season
- * Rainfall
- * Temperature
- * Population or number of water connections
- * Previous supply levels
- * Holidays or special events

A machine learning model will analyze historical patterns and predict future water demand for each selected zone.

Possible models include Random Forest, XGBoost, Support Vector Regression (SVR), and, if sufficient time and data are available, LSTM.

The models will be evaluated using suitable forecasting metrics, and the best-performing model will be selected.

The predicted demand can then be used to support water distribution planning. For example, if one zone is predicted to have significantly higher demand than another zone, planners can consider this information when deciding how to allocate available water.

The system will present predictions through a simple dashboard so that a non-technical planner can easily understand the expected water demand in different zones.

3. Data

The project will require historical water demand data for selected urban zones.

Potential variables include:

- * Zone or area
- * Date
- * Water demand
- * Water supplied
- * Number of water connections
- * Population estimate
- * Rainfall
- * Temperature
- * Day of the week
- * Month or season
- * Holiday or special-event indicator

Weather information may be obtained from publicly available weather datasets, while population information could come from publicly available demographic sources.

The project will prioritize datasets that are consistent, sufficiently large, and suitable for machine-learning experimentation.

Open question about data

Can sufficiently detailed and consistent water demand data be obtained for selected urban zones within the capstone timeline?

4. Success Criteria

The project will be considered successful if:

1. The system can accept and process historical water demand data.
2. The machine learning model can predict future water demand with measurable performance on an appropriate test dataset.
3. The system can predict demand for selected urban zones.
4. The system can compare predicted demand between different zones.
5. The predictions can support water distribution planning.
6. The system can display predictions clearly through a dashboard.
7. The system can provide understandable information about the major factors affecting the prediction where appropriate.
8. The application is usable by a non-technical water planning user.
9. The complete system can be demonstrated using realistic urban water supply scenarios.

Model performance will be evaluated using appropriate regression or forecasting metrics such as MAE, RMSE, and R^2 . The final metrics will depend on the characteristics of the selected dataset and the final forecasting approach.

5. Scope Cut

The initial version will not attempt to predict water demand for every urban area in Rwanda or directly control physical water infrastructure.

The project will focus on a limited number of selected urban zones and a manageable forecasting period.

The system will provide demand predictions and planning support rather than automatically controlling pumps, pipelines, reservoirs, or other physical infrastructure.

This scope limitation is necessary because obtaining nationwide water demand data and integrating the system directly with physical water infrastructure would significantly increase the project's complexity.

TASK 6: REFLECTION

1. How did the idea change between Task 1 and Task 4?

The idea changed from a general water demand prediction concept into a more specific AI-Based Water Distribution Demand Prediction System.

During the brainstorming process, I realized that predicting water demand for different urban zones would provide useful information for water distribution planning.

The main change was therefore making the system focus on predicting future water demand for selected urban zones and presenting the predictions in a way that can support distribution decisions.

I also reduced the scope from a broad water distribution system to a limited number of selected urban zones. This makes the project more realistic and achievable within the capstone timeline.

2. What is the part of the proposal I am least confident about?

I am least confident about the availability and quality of historical water demand data for different urban zones.

The machine learning model requires enough historical data to identify reliable demand patterns. Additional information such as rainfall, temperature, population, and the number of water connections may also be difficult to obtain in a consistent format.

To become more confident, I need to identify the exact datasets available for the selected zones, examine their completeness and consistency, and test whether they contain enough historical information to support reliable water demand prediction.